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# Spectral Analysis Methodology

How we measure what a star is made of, starting with the Sun

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METHODOLOGY

# Spectral Analysis Methodology

How we measure what a star is made of, step by step, starting with the Sun

SECTION 01

## One star, A to Z

The question is simple to state. How much of each element is in a star's outer layers? Every element absorbs light at its own set of wavelengths. Those absorption lines show up as dark notches in the star's spectrum, the star's light spread out by wavelength. The depth and shape of each notch tell us how many atoms of that element are there.

We start with the Sun. Its composition has been measured carefully by other teams, so it is the test. If our method cannot reproduce the published solar values, it has no business measuring other stars. The same process will then be applied to stars that host planets.

### THE SIXTEEN STEPS

Every star and every element goes through the same steps, in this order. Steps 1 to 8 are preparation. Nothing is measured until they are done.

1 **Choose the star.**

2 **Find every instrument and data holding.** A holding is one set of observations of the star from one instrument, as we store it.

3 **Audit each holding.** Check what the files really are: their units, their wavelength scale and whether anything has already been done to them.

4 **Correct telluric absorption and normalise the continuum, once per holding, then freeze.** The prepared spectrum is then used unchanged for every element.

5 **Collect the literature on the star.** What have other teams measured, and how?

6 **Secure the atomic line data.** Wavelengths and line strengths for every line we might use.

7 **Grade lines against the published solar sets.**

8 **Fix or solve the stellar parameters.** Temperature, surface gravity and the other numbers that describe the star's atmosphere.

9 **Measure iron first.** Iron has more usable lines than any other element, so it is the first check that everything above is working.

10 **Measure every band, instrument and engine separately.**

11 **Uncertainty analysis.** Build the error bar for each result.

12 **Check against the literature.**

13 **Report problem lines.** Lines that disagree with the rest are listed, not quietly dropped.

14 **Derived ratios.** Ratios such as carbon to oxygen, which matter for planets.

15 **Human review.**

16 **Publication.**

THE ORCHESTRATOR

The steps are run by one program, the orchestrator. We give it one command per star and element. It works through every combination of band, instrument and calculation method, and reports the status of each one: done, held back and why, or not run yet.

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SECTION 02

## Data

No single instrument covers every wavelength we need, so we use several. Each one is measured on its own. Having more than one look at the same lines also shows us when a result depends on the instrument rather than on the star.

### WAVELENGTH BANDS

We split the spectrum into bands. Each band is handled under its own rules, because the problems change with wavelength. In the near-UV, lines are packed so tightly that no line stands alone. In the red and infrared, lines are sparse but Earth's atmosphere absorbs strongly.

| BAND | WAVELENGTHS (Å) |
| --- | --- |
| Near-UV | 3000 – 3780 |

| BAND | WAVELENGTHS (Å) |
|---|---|
| VIS (visible) | 3780 – 6910 |
| Red-optical | 6910 – 9199 |
| NIR (near-infrared) | 9199 – 13000 |
| H (infrared) | 15007 – 17494 |
| K (infrared) | 19452 – 24846 |

## SOLAR HOLDINGS BY BAND

| HOLDING | WHAT IT IS | BANDS USED |
|---|---|---|
| Kitt Peak solar atlas, our telluric correction | The 1984 Kitt Peak spectrum of sunlight, corrected for Earth's atmosphere by us with molecfit | Near-UV, VIS, red-optical, NIR |
| Kitt Peak solar atlas, Kurucz 2005 correction | The same atlas with Earth's atmosphere removed by Kurucz (2005) | Near-UV, VIS, red-optical, NIR |
| IAG solar atlas | A second, independent spectrum of sunlight from Göttingen, published already corrected (Reiners et al. 2016; Baker et al. 2020) | VIS, red-optical, NIR |

| HOLDING | WHAT IT IS | BANDS USED |
|---|---|---|
| HARPS | Ten direct exposures of the Sun from the HARPS spectrograph in Chile, corrected by us with molecfit | VIS |
| CRIRES+ | Infrared spectra of sunlight reflected off the asteroid Vesta, from the CRIRES+ spectrograph in Chile | NIR, H, K |

Raw spectra are never used. Every spectrum in this table has been corrected for Earth's atmosphere and normalised before any line is measured.

### OTHER TARGETS

We also hold spectra of planet hosts and nearby stars, including 55 Cancri A and α Centauri A and B. These include HARPS, ESPRESSO, CRIRES+ and Hubble ultraviolet spectra. They wait until the Sun is finished. Each system page lists what we have for it.

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SECTION 03

## Telluric correction and continuum

Before a spectrum can be measured, two things have to be removed: the fingerprint of Earth's air, and the overall brightness level. We do this once for each holding. The result

is then frozen and used for every element, so all elements see the same spectrum.

## TELLURIC LINES

Telluric lines are absorption by Earth's own atmosphere, mostly water vapour and oxygen. They land on top of the star's lines and can make a line look deeper than it is. Every spectrum taken from the ground is corrected. Only a telescope in space escapes them.

For HARPS and CRIRES+ we model the atmosphere on the night of each observation with molecfit, a tool from the European Southern Observatory (Smette et al. 2015). The Kitt Peak atlas has no record of when or where each part was observed, so we fit its atmosphere freely. We also use the independent Kurucz (2005) correction of the same atlas. The IAG atlas comes already corrected by its authors.

## CONTINUUM

The continuum is the brightness the star would have with no absorption lines at all. Line depths are measured against it, so it has to be set to 1 first. This is called normalising.

Published solar studies place the continuum at the local high points of the spectrum near each line. We do the equivalent inside the fit. Near each line we compare the observed spectrum with our own calculated spectrum, and set the continuum level from that comparison. Points that disagree badly, for example because of an unlisted line, are clipped out first.

For weak lines this matters a great deal. A line only 1% deep is changed noticeably by a continuum error of a tenth of a percent. So we also repeat each measurement on the second, independent IAG atlas. The difference goes into the error bar as the continuum uncertainty.

In the near-UV the true continuum is never seen, because lines cover every wavelength. There we can only place a pseudo-continuum, and its uncertainty is priced separately.

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SECTION 04

## Atomic data and line grades

To turn a line's depth into a number of atoms, we need to know how strongly a single atom absorbs at that wavelength. This is the line's oscillator strength, written gf. A wrong gf gives a wrong abundance, however good the spectrum is. So we choose lines by how well their gf is known.

### THREE LINE GRADES

| GRADE | WHICH LINES | WHY |
|---|---|---|
| Reference | The lines used by the published solar analysis, with the gf values those authors used. That is Asplund, Amarsi & Grevesse (2021), or the paper they cite for that element. Any line with a laboratory-measured gf is added. | We repeat published work first, then extend it. If we measure the same lines with the same atomic data, any difference from the published answer comes from our spectra or our method, and we can find it. |

| GRADE | WHICH LINES | WHY |
|---|---|---|
| Codex | Any line whose gf comes with a published uncertainty: a laboratory measurement, a stated error, or a NIST accuracy grade of A to C. Lines up to 60% deep. | A wider set of lines, each with an error we can put a number on. |
| Deep | The same gf standard as Codex, for lines deeper than 60%. | Strong lines are more sensitive to the star's gas motions and to line broadening. They are reported, but as a secondary result. |

A line whose gf has no stated uncertainty can still be measured, but it does not reach a grade. We note the gap rather than invent an error for it.

## ONE GF PER PHYSICAL LINE

Many lines are really a cluster of very close components, split by the structure of the atom's nucleus (hyperfine structure) or by its isotopes. The calculation needs each component to get the line's shape right. But the gf is fixed once for the whole physical line, from a single master table. The components are scaled together so that their total matches it. A line with no entry in that table stops the run with an error. Nothing falls back to a default.

### LINES WE REMOVE

Published solar studies leave out lines that are saturated, meaning so strong that the depth barely changes with abundance, and lines too blended with neighbours to measure. We do the same. A saturated line that the published solar analyses did not use is proposed for removal, and a person approves each one. Blends are handled by calculating every known atomic and molecular line in the window together.

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SECTION 05

## Model atmospheres and NLTE

To predict what a line should look like, we need a model of the star's outer layers: how temperature and pressure change with depth. This is a model atmosphere. We also need to know how the atoms in it behave. Different choices give slightly different answers, so we run several and report each one on its own.

### MODEL ATMOSPHERES

A one-dimensional (1D) model treats the atmosphere as flat layers, each with one temperature. We use two standard 1D families: MARCS (Gustafsson et al. 2008) and ATLAS9 (Castelli & Kurucz 2003). The real Sun has rising hot gas and sinking cool gas side by side. A ⟨3D⟩ model is built from a full 3D simulation of that motion (the Stagger grid, Magic et al. 2013), averaged into layers. Swapping MARCS for ATLAS9 on the same lines is one of the measured terms in the error bar.

## NLTE

The simplest calculation assumes LTE, local thermodynamic equilibrium: that each layer's atoms are set by its temperature alone. In a star's thin outer layers that is not quite true. Light from deeper down also excites the atoms. Allowing for this is called NLTE (non-LTE). It can shift an abundance by a few hundredths of a dex or more. (Abundances are on a log scale, and one dex is a factor of ten, so 0.01 dex is about 2%.) We include NLTE in two ways. The first takes published correction tables for each line (from Amarsi, Bergemann and collaborators) and applies them to the result. Where those tables stop, we compute corrections the same way from the published model atoms (Amarsi et al. 2020). The second puts NLTE inside the calculation itself. Turbospectrum (Gerber et al. 2023) reads departure coefficients, which say how far each atomic level is from LTE, and builds them into the predicted spectrum. That is done on both MARCS 1D and ⟨3D⟩ models.

| TREATMENT | ATMOSPHERE | HOW NLTE ENTERS |
|---|---|---|
| 1D LTE | ATLAS9 or MARCS | None |
| 1D NLTE, correction tables | ATLAS9 | Per-line correction from published tables |
| 3D NLTE, correction tables | ATLAS9 base | Per-line 3D NLTE correction from published tables |
| 1D NLTE, in the calculation | MARCS | Departure coefficients inside the synthesis |

| TREATMENT | ATMOSPHERE | HOW NLTE ENTERS |
|---|---|---|
| ⟨3D⟩ LTE and ⟨3D⟩ NLTE | Stagger ⟨3D⟩ | None, then departure coefficients, on the same atmosphere |

Each treatment is its own result. They are never averaged together. If two treatments disagree, the disagreement is information about the physics, and averaging would hide it.

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SECTION 06

## Measuring an abundance

An abundance is the number of atoms of an element for every trillion hydrogen atoms, on a log scale. For the Sun, iron is about 7.46 on that scale (Asplund et al. 2021). We find it by changing the abundance in our calculation until the predicted line matches the observed one.

### SPECTRUM SYNTHESIS, THE MAIN ROUTE

Spectrum synthesis means calculating the whole stretch of spectrum around a line: the line itself, its neighbours, and any molecules there. We use Turbospectrum (Plez 2012) through iSpec (Blanco-Cuaresma et al. 2014). The program changes the abundance until the calculated spectrum best fits the observed one. Because every neighbour is in the calculation, a blend is modelled rather than ignored. In the near-UV this is the only route allowed, since no line there stands alone.

### PROFILE FIT, THE SECOND ROUTE

In the visible and red bands, many lines are well separated. There we can also fit the line's own shape and measure its equivalent width: the width of a fully dark strip with the same area as the line. A curve of growth, which links equivalent width to abundance, turns it into an abundance. The two routes are kept as separate results. Simply adding up the missing light over a fixed window is not used. Tested against the HARPS lines, it clipped the wings of strong lines and took in neighbours of weak ones.

### FROM LINES TO ONE NUMBER

Each line gives its own abundance. A result combines the lines of one element, from one band, one instrument holding and one treatment. Its value is the median over those lines, so a single odd line cannot pull it far. When the published analysis we are repeating used a plain average, we use the same.

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SECTION 07

## Uncertainty

A number without an honest error bar cannot be compared with anything. Each result has two layers. The error bar we show is worked out the way Asplund, Amarsi & Grevesse (2021) work out theirs, so our numbers and theirs can be compared directly.

Behind it sits a much longer check of everything else that could move the answer. A result that fails that check is not published at all.

## THE ERROR BAR WE SHOW

It has three parts, added in quadrature (the square root of the sum of the squares):

| PART | WHAT IT IS |
|---|---|
| Statistical | How much the lines disagree with each other: their scatter divided by the square root of their number. With one or two lines there is no scatter to measure, so we use how sharply the fit gets worse as the abundance moves away from the best value. |
| 3D model | Half of how much the answer changes between model atmospheres, including the change from a simple layered model to a full 3D model of the Sun's churning surface. |
| NLTE | Half of the NLTE correction, and never less than 0.03 dex, even where the correction is small or not calculated. |

As in Asplund et al. (2021), errors in the atomic data (gf values) are not part of this bar. They are measured and checked in the longer list below.

## MOLECULES AND THE 3D MODEL

Carbon, nitrogen and oxygen are also measured from molecules such as CH, CN, CO and OH. Molecules do not need an NLTE term. Instead, a molecule's strength depends

on both of its elements, so the uncertainty in the other element takes that place (Amarsi et al. 2021).

Asplund and Amarsi calculated these molecular lines in full 3D, using simulations of the solar surface and their own software, which is not public. Our programs work with layered (1D) models, so we cannot repeat that calculation. They did publish how far the answer moves between 3D and 1D for each band of each molecule, measured on the same lines we use. We take that published difference for the same band: CH in the G band moves up 0.066 dex, CO near 2.3 microns moves down 0.142, CN in the near-infrared down 0.060. It does two things. Half of it goes into the 3D part of the error bar, which is never smaller than the spread Amarsi found across their four model atmospheres. And when results are combined into one number (below), it puts our 1D result on the same 3D footing as theirs. The result's own measured value is never changed. The page shows both.

## COMBINING RESULTS INTO ONE NUMBER

For carbon, nitrogen and oxygen the top number combines the different kinds of measurement, as Asplund et al. (2021) do. Each kind is represented by our best result for it, on the 3D footing described above.

Carbon and oxygen take a weighted average, where a result with a smaller error bar counts for more. The statistical parts shrink as results are added, because they are independent. The model parts do not shrink, because every result shares the same model of the Sun (Amarsi et al. 2021).

Nitrogen takes the plain average of the atomic lines and the molecules. Its error bar is half the gap between those two, or their own error bars combined, whichever is larger. When the two agree closely, half the gap alone would claim more precision than either measurement has.

One term Amarsi et al. (2021) include for molecules is not in our bar yet: how much the answer drifts with a line's wavelength or strength across a band. Measuring it needs a

separate answer for each line, and our molecular fits give one answer for the whole band.

## THE FULL CHECK BEHIND EACH RESULT

For every term below we re-run the same lines with one thing changed and record how far the answer moves. Each result has to carry every term. If any one is missing, the result is held back and nothing is published.

| TERM | WHAT WE CHANGE OR CHECK |
|---|---|
| Atomic data (gf) | The published gf uncertainties, combined as explained below |
| Temperature | The star's temperature |
| Surface gravity | The star's surface gravity |
| Metallicity | The star's overall metal content |
| Microturbulence | The small-scale gas motion that broadens strong lines, raised and lowered |
| Continuum | Two placements of the continuum, and the second, independent IAG atlas |
| Fit window | The width of the stretch of spectrum fitted |
| Pseudo-continuum | Near-UV only, where the true continuum is never seen |

| TERM | WHAT WE CHANGE OR CHECK |
|---|---|
| Telluric | What is left of Earth's atmosphere after correction, measured against an independent correction |
| Method check | How far the method misses the known answer in the visible band |
| NLTE | The uncertainty in the NLTE correction |
| Model atmosphere | MARCS swapped for ATLAS9 |
| Hyperfine and isotopes | The splitting of a line into close components |
| Blends | Neighbouring lines missing from the calculation |
| Molecules | For a molecular line, its dependence on the other element in the molecule |

Some terms do not apply to every result. Pseudo-continuum is zero outside the near-UV, for example. For the Sun, temperature, gravity and metal content are known so well that those terms come out at zero. Microturbulence is not, and is measured. The full list is kept with every result and can be downloaded with it.

## GF ERRORS THAT MOVE TOGETHER

Laboratory gf values usually come from two measurements: how long an excited atom lives, and how its decay splits between lines. The lifetime sets an overall scale shared by every line from that source. If it is off, all those lines are off in the same direction, and

averaging more of them does not help. The other part of each line's error is its own, and does average down.

So for each gf source we record which kind of error it carries. The shared scale is counted once. The per-line part shrinks with the number of lines. A source we have not yet reviewed is treated the cautious way, as fully shared.

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SECTION 08

## Publication and solar validation

A result is not published because it was computed. It has to pass a comparison with the literature and a human review first. Then it goes on the site, and anything it replaces is archived.

### CHECKING AGAINST THE PUBLISHED SUN

For every element we collect the published solar values first, chiefly Asplund, Amarsi & Grevesse (2021) and Amarsi & Grevesse (2026), each with its method. Our result is compared with them. The visible band is the control. It has a known answer, iron at 7.46 (Asplund et al. 2021). A method that cannot reproduce it there has not earned the right to report a value in a band with no known answer. How far it misses is measured and goes into the error bar as the method check.

### PROBLEM LINES AND HUMAN REVIEW

Lines that disagree with the rest, or fail a check, are reported for each run with the reason. Every proposed line removal, and every publication, is approved by a person. The program does the measuring. A person decides what is published.

**ONLY CURRENT RESULTS COUNT**

When a result is re-measured, for example after a correction to the atomic data, the new one replaces the old. The old one is moved to an archive with the record of why. Nothing is deleted, so any earlier number on this site can be traced.

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SECTION 09

## References

The full list, with links, is on the Sources page.

| REFERENCE | PUBLICATION |
|---|---|
| Amarsi & Grevesse (2026) | arXiv:2608.23155 — Solar chemical composition |
| Amarsi et al. (2019) | A&A 630, A104 — Carbon NLTE corrections |
| Amarsi et al. (2020) | A&A 642, A62 — NLTE departure coefficients for large surveys |
| Amarsi et al. (2021) | A&A 656, A113 — Solar C, N and O from molecular lines |
| Amarsi et al. (2022) | A&A 668, A68 — Iron NLTE corrections |
| Asplund, Amarsi & Grevesse (2021) | A&A 653, A141 — The chemical make-up of the Sun: a 2020 vision |

| REFERENCE | PUBLICATION |
| --- | --- |
| Baker, Blake & Reiners (2020) | ApJS 247, 24 — The IAG solar flux atlas: telluric correction |
| Barklem & Aspelund-Johansson (2005) | A&A 435, 373 — Van der Waals damping constants |
| Bergemann et al. (2012) | MNRAS 427, 27 — Iron NLTE in 1D and ⟨3D⟩ model atmospheres |
| Bergemann et al. (2017) | ApJ 847, 15 — NLTE with 1D and ⟨3D⟩ models: magnesium |
| Blanco-Cuaresma et al. (2014) | A&A 569, A111 — iSpec spectral analysis framework |
| Castelli & Kurucz (2003) | Proc. IAU Symp. 210 — ATLAS9 model atmosphere grid |
| Gerber et al. (2023) | A&A 669, A43 — NLTE radiative transfer with Turbospectrum |
| Gustafsson et al. (2008) | A&A 486, 951 — MARCS model atmosphere grid |
| Kramida et al. (2023) | NIST Atomic Spectra Database — physics.nist.gov/asd |
| Kurucz (2005) | High-resolution solar irradiance and telluric-corrected solar atlas |
| Lodders, Bergemann & Palme (2025) | Space Sci. Rev. 221 — Solar system abundances from the photosphere and CI chondrites |

| REFERENCE | PUBLICATION |
| --- | --- |
| Magic et al. (2013) | A&A 560, A8 — The Stagger grid: averaged 3D model atmospheres |
| Pehlivan Rhodin et al. (2017) | A&A 598, A102 — Oscillator strengths of Mg I |
| Pehlivan Rhodin et al. (2024) | A&A 682, A184 — Transition data for Si I and Si II |
| Plez (2012) | Turbospectrum: Code for spectral synthesis — radiative transfer, native molecular lines |
| Reiners et al. (2016) | A&A 587, A65 — The IAG solar flux atlas |
| Ryabchikova et al. (2015) | PhyS 90, 054005 — VALD3 atomic line database |
| Schmitt (2010) | Senior Thesis, University of Montana — predecessor 55 Cnc abundance study |
| Smette et al. (2015) | A&A 576, A77 — Molecfit: telluric absorption correction |


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